جامعة الإمارات العربية المتحدة
United Arab Emirates University

College of Information Technology

Inventory Management System (MVC)

Internet Computing (01)

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Term Project

Group 1

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1. Domain Description & System Overview

The Inventory Management System is a web-based application aimed at simplifying product and supplier management processes in a retail or small business setup. The system allows users to do basic inventory operations like adding new products, changing existing product details, deleting old records, and checking the total stock.

Moreover, the system offers supplier management capabilities to the user, enabling the user to keep up with the supplier records such as names, contact numbers, and emails.

We have included a simple authentication module in this system to be sure that only registered users are allowed to use the system's functions.

This is a CRUD (create, read, update, delete) system project that is fully implemented with Java Servlets, JSP, JSTL, DAO design, and is based on the MVC architecture. The backend database is in MySQL and accessed through MAMP.

2. MVC Architecture Overview

The system is structured using the Model–View–Controller (MVC) architectural pattern:

Model (M): Represents the data and the business logic.

- Product Class
- Supplier Class
- User Class
- DAO Classes (ProductDAO, SupplierDAO, UserDAO)

The model is responsible for storing object data, validating input, and querying/updating the database.

View (V): Includes all JSP pages responsible for displaying the user interface.

- index.jsp (First page before login)
- home.jsp (Dashboard after login)
- login.jsp
- register.jsp
- error.jsp
- listProducts.jsp
- listSuppliers.jsp
- add_product.jsp
- add_supplier.jsp
- update_product.jsp
- update_supplier.jsp

These pages define the HTML structure, CSS styling, and form layouts.

Controller (C): Implemented using Java Servlets they receive user requests, process inputs, call the appropriate DAO methods, and redirect the user back to JSP pages.

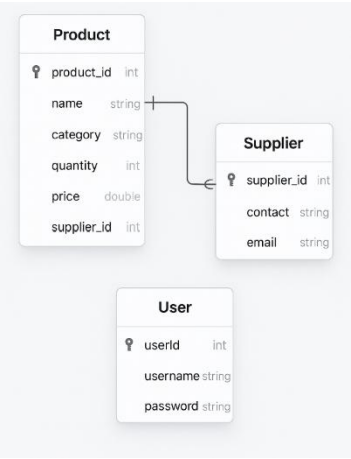
Main Controllers:

- LoginServlet
- RegisterServlet
- LogoutServlet
- AddProductServlet
- UpdateProductServlet
- DeleteProductServlet
- ListProductServlet
- AddSupplierServlet
- UpdateSupplierServlet
- DeleteSupplierServlet
- ListSupplierServlet

The MVC structure ensures separation of concerns, easier debugging, and clean maintainability.

3. Database ERD (Entity-Relationship Diagram)

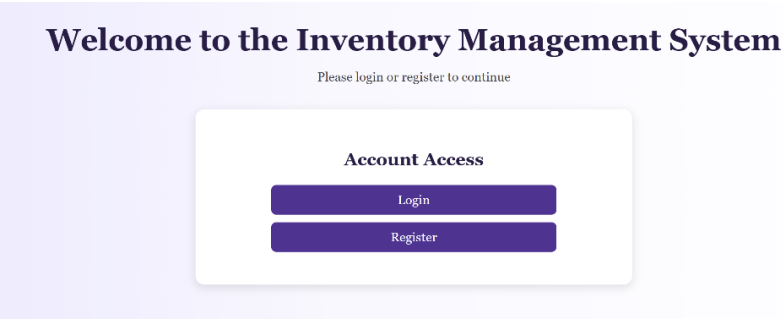
The database designed for the Inventory Management System aims to be straightforward, scalable, and completely in line with the MVC structure of the app. Basically, it covers the three core entities: User, Supplier, and Product. The User entity is a standalone one and keeps the authentication data for login and registration. The Supplier and Product entities are connected by a one-to-many relationship, thus a supplier can have many products, but each product is linked to only one supplier. Such a layout preserves referential integrity and makes CRUD operations throughout the system efficient. The ERD below helps to understand the entities and their features as well as the relationships between them.



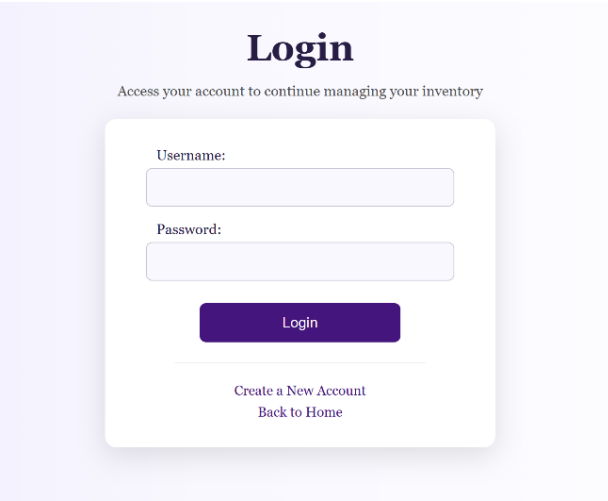
4. System Implementation Summary

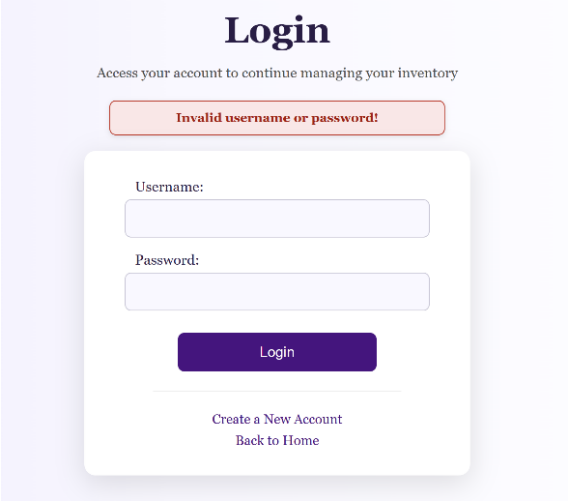
The front end of the Inventory Management System was achieved with JSP pages that were given a fresh look using modern CSS to maintain clarity, consistency, and user-friendliness. Every page is an integral part of the system's working process, escorting the user from logging in, handling products, managing suppliers, through to performing CRUD operations. The subsequent screen grabs exhibit the essential elements of the system, thus the user can see how the app functions under regular conditions along with validation and confirmation instances.

4.1 User Authentication Screens

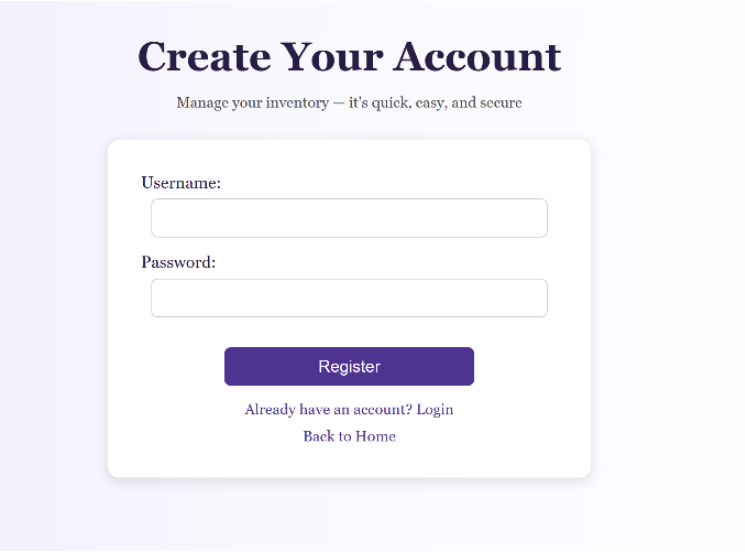


User is prompted to Login or Register.



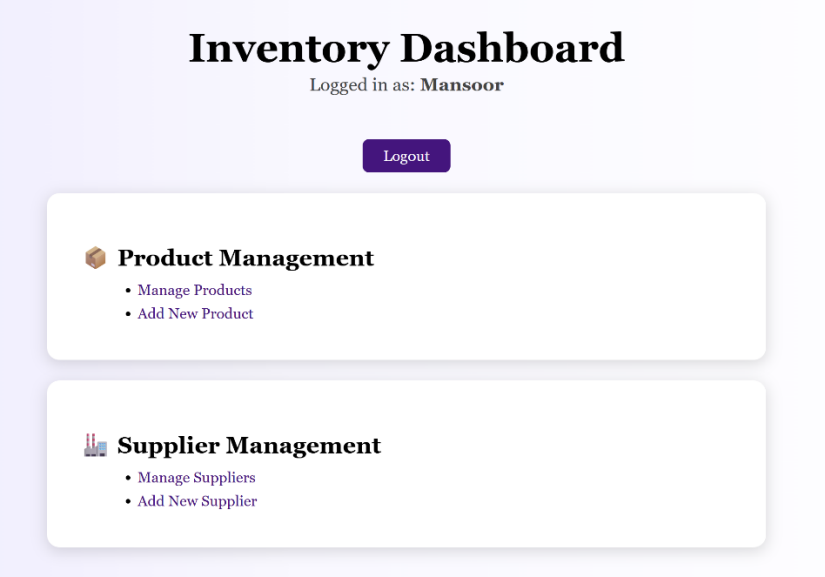


User enters credentials to access the system, and invalid attempts show an error message.



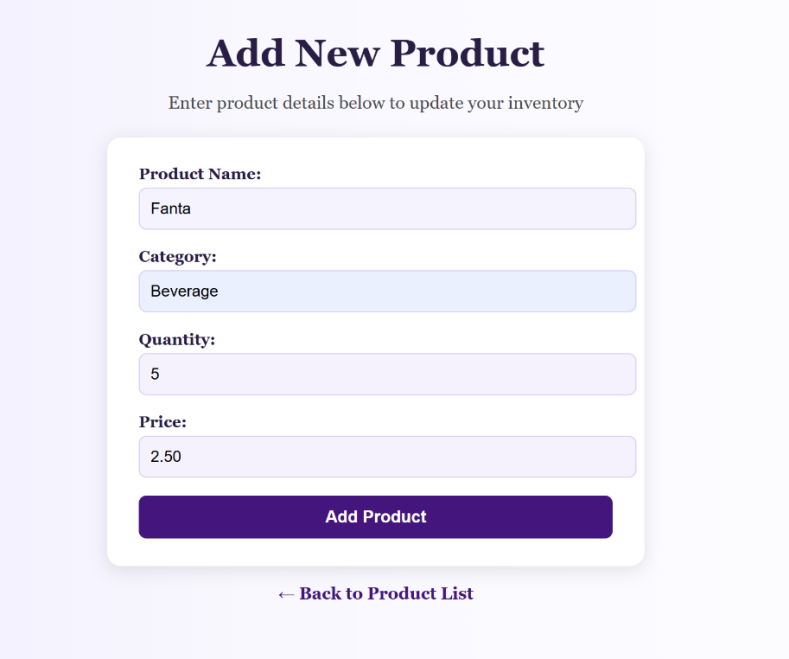
User creates a new account to access the inventory dashboard.

4.2 Home Dashboard

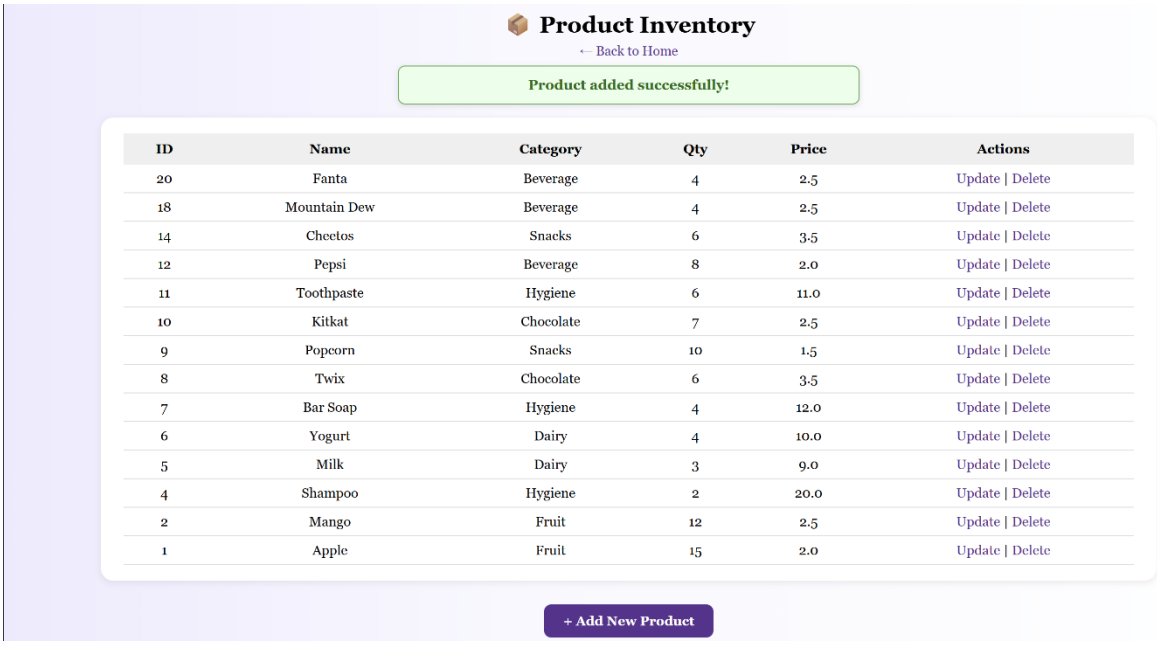


After logging in, the user is redirected to the Home Dashboard, where they can access product and supplier management pages. If a logged-in user closes the site and later reopens it while the session is still active, they will automatically be redirected back to this dashboard.

4.3 Product Management

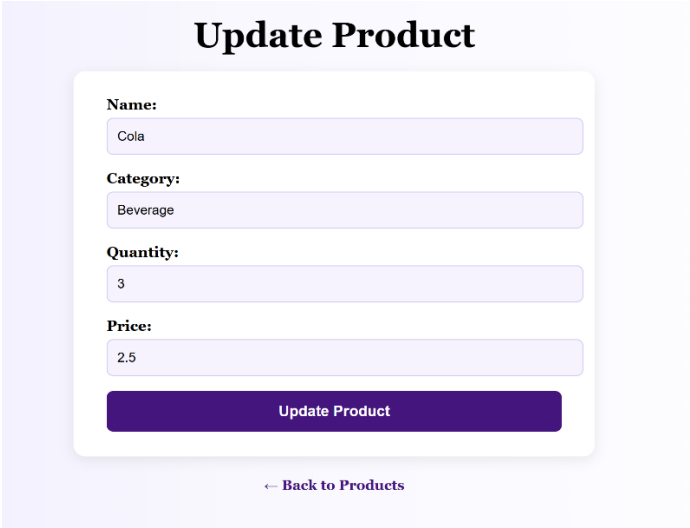


User can insert new product details including name, category, quantity, and price.

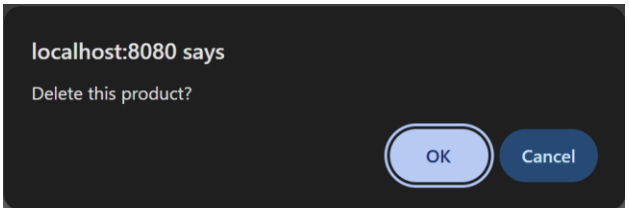


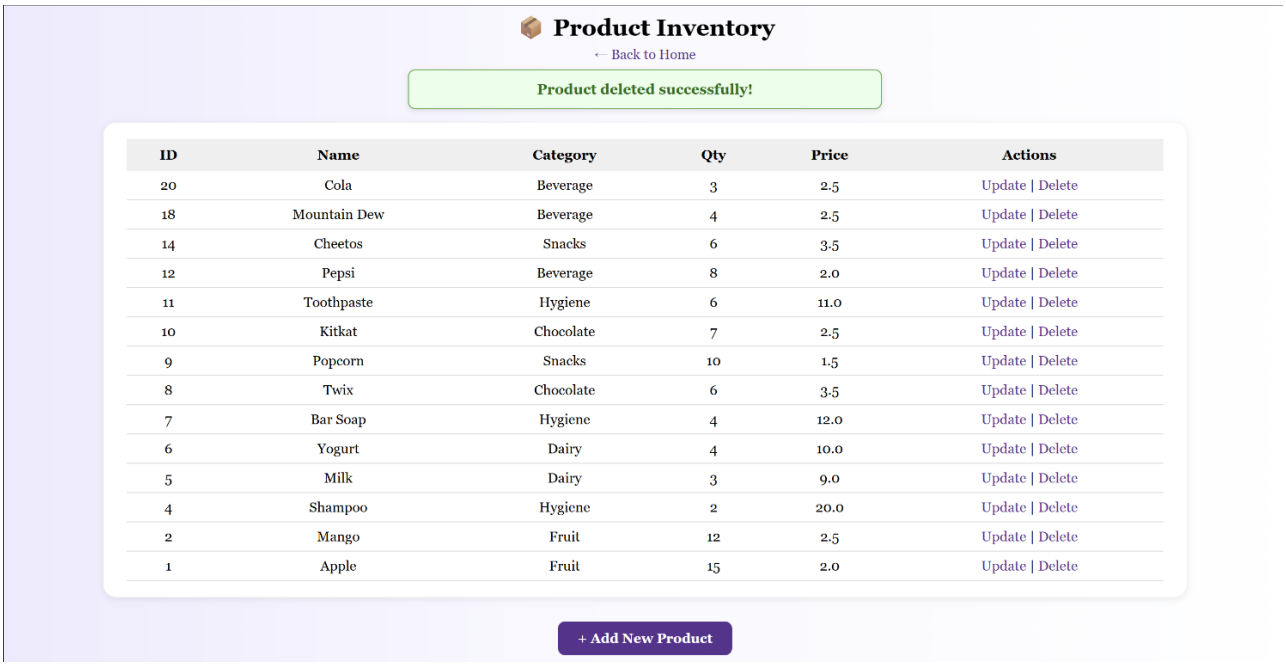
ID	Name	Category	Qty	Price	Actions
20	Fanta	Beverage	4	2.5	Update | Delete
18	Mountain Dew	Beverage	4	2.5	Update | Delete
14	Cheetos	Snacks	6	3.5	Update | Delete
12	Pepsi	Beverage	8	2.0	Update | Delete
11	Toothpaste	Hygiene	6	11.0	Update | Delete
10	Kitkat	Chocolate	7	2.5	Update | Delete
9	Popcorn	Snacks	10	1.5	Update | Delete
8	Twix	Chocolate	6	3.5	Update | Delete
7	Bar Soap	Hygiene	4	12.0	Update | Delete
6	Yogurt	Dairy	4	10.0	Update | Delete
5	Milk	Dairy	3	9.0	Update | Delete
4	Shampoo	Hygiene	2	20.0	Update | Delete
2	Mango	Fruit	12	2.5	Update | Delete
1	Apple	Fruit	15	2.0	Update | Delete

All products are displayed with options to update or delete each item.



Allows editing and saving changes to existing product information.

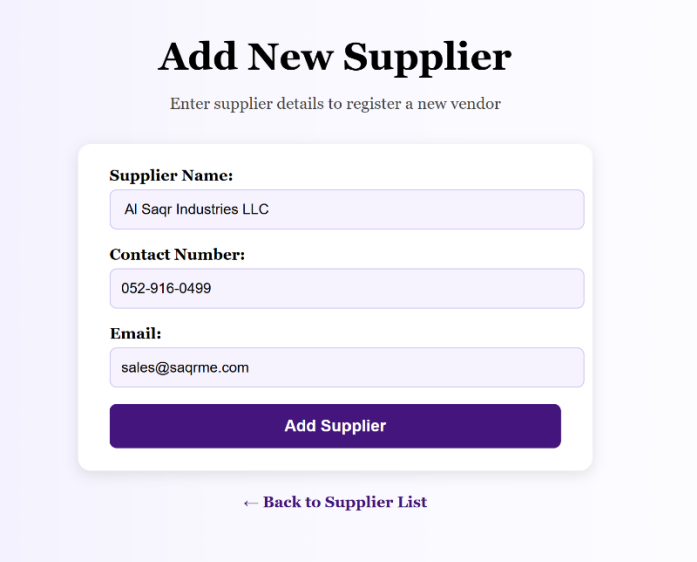




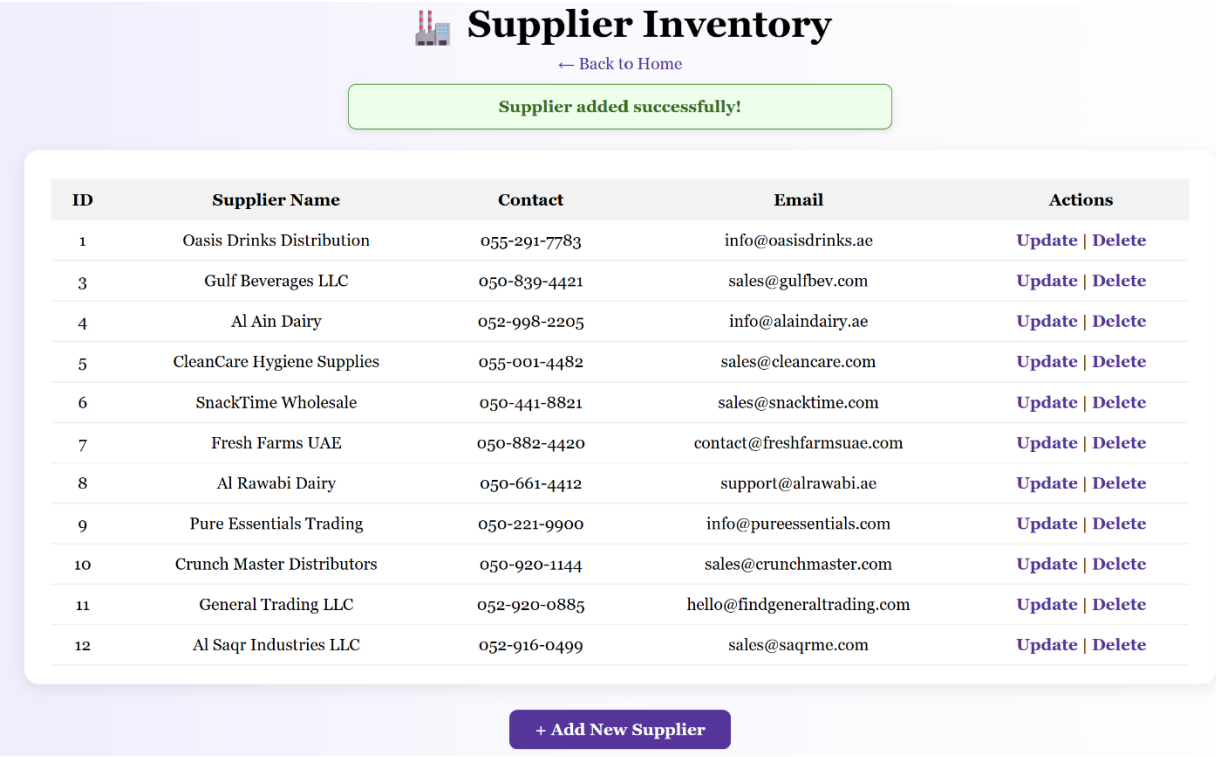
ID	Name	Category	Qty	Price	Actions
20	Cola	Beverage	3	2.5	Update | Delete
18	Mountain Dew	Beverage	4	2.5	Update | Delete
14	Cheetos	Snacks	6	3.5	Update | Delete
12	Pepsi	Beverage	8	2.0	Update | Delete
11	Toothpaste	Hygiene	6	11.0	Update | Delete
10	Kitkat	Chocolate	7	2.5	Update | Delete
9	Popcorn	Snacks	10	1.5	Update | Delete
8	Twix	Chocolate	6	3.5	Update | Delete
7	Bar Soap	Hygiene	4	12.0	Update | Delete
6	Yogurt	Dairy	4	10.0	Update | Delete
5	Milk	Dairy	3	9.0	Update | Delete
4	Shampoo	Hygiene	2	20.0	Update | Delete
2	Mango	Fruit	12	2.5	Update | Delete
1	Apple	Fruit	15	2.0	Update | Delete

A confirmation dialog asks the user to confirm before deleting a product and applies the changes to the table immediately with a success pop-up.

4.4 Supplier Management

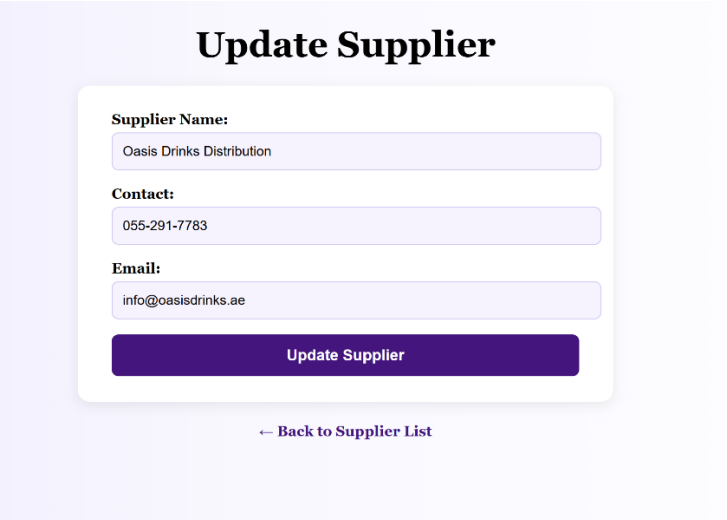


User can add vendor information including name, contact, and email.

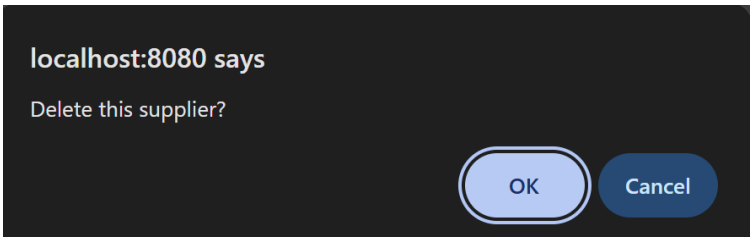


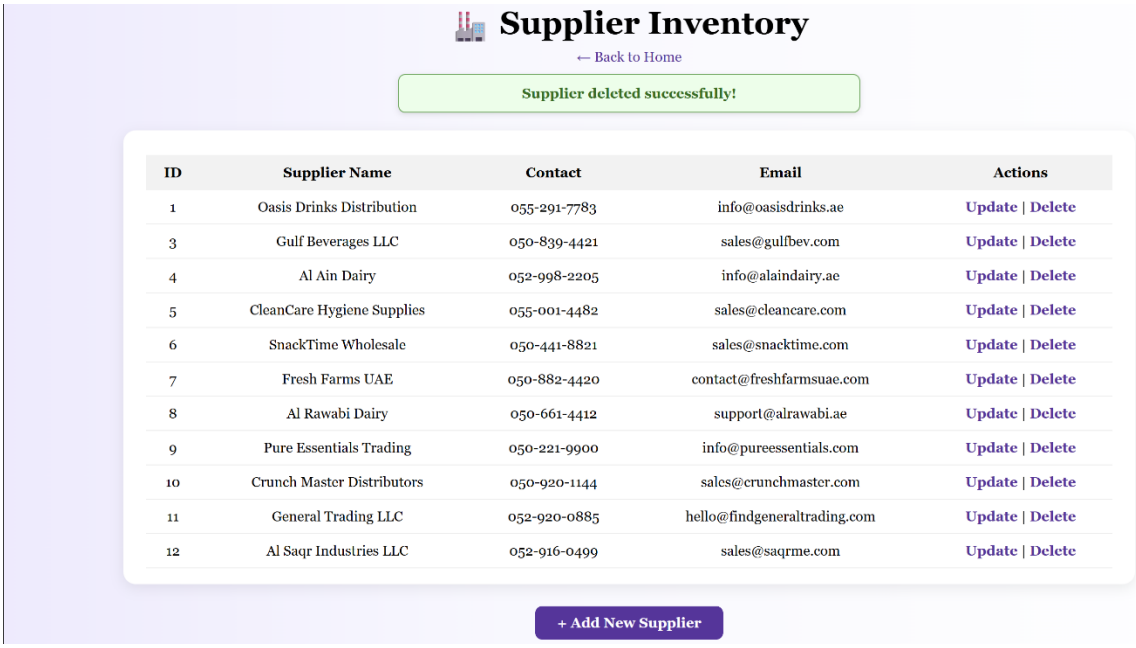
ID	Supplier Name	Contact	Email	Actions
1	Oasis Drinks Distribution	055-291-7783	info@oasisdrinks.ae	Update | Delete
3	Gulf Beverages LLC	050-839-4421	sales@gulfbev.com	Update | Delete
4	Al Ain Dairy	052-998-2205	info@alaindairy.ae	Update | Delete
5	CleanCare Hygiene Supplies	055-001-4482	sales@cleancare.com	Update | Delete
6	SnackTime Wholesale	050-441-8821	sales@snacktime.com	Update | Delete
7	Fresh Farms UAE	050-882-4420	contact@freshfarmsuae.com	Update | Delete
8	Al Rawabi Dairy	050-661-4412	support@alrawabi.ae	Update | Delete
9	Pure Essentials Trading	050-221-9900	info@pureessentials.com	Update | Delete
10	Crunch Master Distributors	050-920-1144	sales@crunchmaster.com	Update | Delete
11	General Trading LLC	052-920-0885	hello@findgeneraltrading.com	Update | Delete
12	Al Saqr Industries LLC	052-916-0499	sales@saqrme.com	Update | Delete

Displays all suppliers with options to update or delete.



Allows editing and updating supplier records,





ID	Supplier Name	Contact	Email	Actions
1	Oasis Drinks Distribution	055-291-7783	info@oasisdrinks.ae	Update | Delete
3	Gulf Beverages LLC	050-839-4421	sales@gulfbev.com	Update | Delete
4	Al Ain Dairy	052-998-2205	info@alaindairy.ae	Update | Delete
5	CleanCare Hygiene Supplies	055-001-4482	sales@cleancare.com	Update | Delete
6	SnackTime Wholesale	050-441-8821	sales@snacktime.com	Update | Delete
7	Fresh Farms UAE	050-882-4420	contact@freshfarmsuae.com	Update | Delete
8	Al Rawabi Dairy	050-661-4412	support@alrawabi.ae	Update | Delete
9	Pure Essentials Trading	050-221-9900	info@pureessentials.com	Update | Delete
10	Crunch Master Distributors	050-920-1144	sales@crunchmaster.com	Update | Delete
11	General Trading LLC	052-920-0885	hello@findgeneraltrading.com	Update | Delete
12	Al Saqr Industries LLC	052-916-0499	sales@saqrme.com	Update | Delete

A confirmation dialog asks the user to confirm before deleting a supplier, and applies the changes to the table immediately.

5. System Implementation

The section explains the Inventory Management System development with Java Servlets, JSP, DAO pattern, and MySQL relational database. The Implementation is based on MVC architecture, which separates the app into Model, View, and Controller layers for better clarity, maintainability, and scalability.

5.1 Model Implementation

The Model layer contains the Java classes that represent the system's main data entities:

- User – stores cedentials for authentication.
- Product – stores product information such as name, category, quality, and price.
- Supplier – stores supplier details including name, contact, and email.

Each class includes private attributes with public getters and setters, allowing safe and structured data manipulation. These objects act as data carriers between the database and application.

5.2 Data Access Layer (DAO) Implementation

The DAO layer is responsible for database communication.

This includes:

- ProductDAO: Handles all product-related operations.
    - Insert new products
    - Retreive list of products
    - Update prodct details
    - Delete products
- SupplierDAO: Handles all supplier-related operations.
    - Insert new suppliers
    - Retreive list of suppliers
    - Update supplier details
    - Delete suppliers
- UserDAO: Handles the authentication-related operations.
    - Registering a new user
    - Validating login credentials
- DBUtil: Although not a DAO itself, DBUtil provides the shared database connection method so that all DAOs can connect to MySQL efficiently.

DAO classes use PreparedStatement to prevent SQL injection and ensure safe queries. All database logic is isolated here to maintain MVC separation.

5.3 Controller Layers (Servlets)

Servlets act as controllers that process HTTP requests, communicate with the DAO layer, and decide which JSP page to display.

Main Servlets:

- LoginServlet - validates user credentials using UserDAO and creates the session
- RegisterServlet - inserts a new user record
- LogoutServlet – invalidates the session
- AddProductServlet, UpdateProductServlet, DeleteProductServlet – perform product CRUD operations
- AddSupplierServlet, UpdateSupplierServlet, DeleteSupplierServlet – perform supplier CRUD operations
- ListProductServlet and ListSupplierServlet - retrieve all records and forward them to JSP pages

Each servlet uses request parameters, creates or updates model objects, calls DAO methods, then forwards or redirects to the appropriate JSP. Session management is used to ensure only logged-in users can access system features.

5.4 View Layer (JSP Interfaces)

JSP pages serve as the presentation layer. They display results returned by the controller and show success or error messages stored in session attributes.

The UI includes:

- Authentication Screens: Login, Registeration, Invalid Login
- Dashboard: Shows navigation options for product and supplier management
- Product Pages: add, update, delete confirmation, and product inventory table
- Supplier Pages: add, update, delete confirmation, and supplier inventory table

JSP pages use embedded Java expressions, session checks, and dynamic table generation to display real-time database information.

5.5 Session Management

Sessions are used to track logged-in users.

The system implements:

- Session creation after successful login.
- Blocking of restricted pages when no session exists.
- Redirecting authenticated users directly to the dashboard if they reopen the site while the session is active.
- Session termination when the user clicks Logout.

This ensures secure and controlled access across all pages.

5.6 Database Implementation

The MySQL relational database was created using three tables:

- User
- Product
- Supplier

Each table contains a primary key, and the products table includes a foreign key (supplier_id) to maintain a one-to-many relationship between suppliers and their products.

(A MySQL dump file is included with the submission, containing the full schema and data.)

6. Challenges and Solutions

The creation of the Inventory Management System has been riddled with challenges at various stages of the project. Each difficulty has deepened our grasp of Java web development and has been instrumental in improving our problem-solving skills. The main challenges and their solutions are summarized below.

6.1 Session Management

One of the major difficulties was making sure that users who have already logged in are not redirected to the login page again. If a user closes the browser tab and opens the website while their session is still valid, the application should take the user directly to the dashboard, thereby avoiding the login page.

Solution:

Session checking is done in index.jsp. In the event that there is a user object in the session, the page goes to home.jsp without waiting. If not, the login and registration options are available for users. This action serves as a smooth user experience without the necessity of having another servlet.

6.2 Safe Deletion (Preventing Accidental Data Loss)

Deleting a product or supplier immediately without a warning could easily lead to mistakes.

Solution:

JavaScript confirmation dialogs were added to both product and supplier list pages. The delete operation only runs if the user clicks OK.

6.3 Consistent Page Styling Across All JSPs

Aligning forms, buttons, tables, and alerts across all pages was challenging, especially since each JSP loads separately.

Solution:

A unified CSS file and consistent HTML structure were applied to all pages, ensuring professional and uniform styling.

7. Conclusion and Reflection

This application successfully illustrates the concept and coding of a full-fledged Inventory Management System with the use of the MVC architecture, Java Servlets, JSP, and MySQL. The system empowers users to register, log in, and manage products and suppliers in a secure and well-organized manner. After the implementation of session management, the program guarantees that only users who have logged in will have access to the dashboard and can carry out inventory transactions.

During the coding of the project, a distinct MVC separation is evident. In the model package, there are definitions of the classes Product, Supplier, and User. The DAO classes undertake the users' authentication-related database operations and the CRUD actions. The controller servlets are in charge of the request handling, they also manage all application logic and finally, they send the data to the JSP views. By this division, the team makes the system more mature, extensible, and less difficult to maintain.

The implemented system is equipped with all the necessary CRUD operations for products and suppliers, in conjunction with the validation messages, session handling, and confirmation prompts. The graphical interface is kept neat and uniform from the user's point of view, and all the pages are consistent with each other, which makes the overall user experience better. An ERD and a well-defined database schema are employed to assist in maintaining the logical relationships between different entities, particularly the one-to-many relationship between suppliers and products.

Constructing this system deepens our knowledge of Java web development, servlet communication, form handling, session control, and database interaction. Besides, it further enhances our capability to design and structure a project in line with professional best practices. This project completion is a great stepping stone toward the future development of more complex systems and web applications.